HACKATHON PROJECT

Student Subject Selection System

Intelligent elective selection using 0/1 Knapsack & Dynamic Programming

OPTIMIZATION | EDUCATION | DAA

PROBLEM DOMAIN

Academic planning • Elective selection

TEAM MEMBERS

1) JASHWANTH
2) MANICHANDRA
3) NARENDAR
4) SHIVA KUMAR

ONE-LINE IDEA

Find the highest-preference combination of electives while respecting credit, workload and mandatory constraints.

The Challenge

REAL-WORLD PROBLEM

Students face multiple elective choices but have limited credits, workload capacity and specific academic preferences.

WHY IT MATTERS

A poor combination can waste available credits, overload the student, or ignore subjects that better match their preferences.

KEY CHALLENGES

- Many possible combinations
- Multiple constraints
- Mandatory subjects
- Need for an explainable result

THE OPTIMIZATION QUESTION

“Which valid combination gives the student the maximum preference score?”

Instead of returning only a final number, the system exposes the selected subjects, totals and constraint status.

Our Innovation

FROM SUBJECT LIST → TO EXPLAINABLE OPTIMAL PLAN

SMART SELECTION

Treat each elective as a 0/1 decision: select it or skip it.

DP OPTIMIZATION

Store the best result for each smaller state and reuse it.

RULE AWARE

Respect credit, workload and mandatory requirements.

EXPLAINABLE OUTPUT

Show selected subjects, score and why the solution is valid.

KEY DIFFERENTIATOR

Not just “recommend a subject” — optimize the entire combination.

The prototype connects a real student decision with a classic DAA optimization technique.

How It Works

INPUT

Student limits
+ subject data

VALIDATE

Check credits,
workload & rules

PROCESS

Build DP states
for each subject

OPTIMIZE

Select / Skip
and keep best

OUTPUT

Subjects + score
+ explanation

USER JOURNEY

1. Enter limits

2. Review electives

3. Run optimization

4. View recommendation

Technology & Algorithm

PYTHON

Algorithm + backend logic

FLASK

Web application backend

SQLITE

Student + subject data

BOOTSTRAP

Responsive interface

ALGORITHM CORE

0/1 KNAPSACK

Each subject is either selected (1) or not selected (0). The objective is to maximize preference value within capacity.

DYNAMIC PROGRAMMING

DP stores optimal subproblem results, avoids repeated calculations and enables final decision reconstruction.

COMPLEXITY **Time: O(n × C)** • **Space: O(n × C) standard DP**

Demo & Impact

SUBJECT SELECTION DASHBOARD

Maximum Credits **7** Maximum Workload **6**

SUBJECT	CR	WORK	SCORE	STATUS
Artificial Intelligence	4	3	90	SELECTED
Cloud Computing	3	2	75	SKIPPED
Data Mining	2	2	60	SELECTED
Cyber Security	4	4	80	SKIPPED

TOTAL **Credits 6 • Workload 5 • Preference Score 150**

RESULT

A feasible combination is produced with a transparent decision trail.

PRACTICAL BENEFIT

Reduces manual comparison of elective combinations.

USER VALUE

Students can see what was selected and why it fits their limits.

Future Vision

CURRENT ACHIEVEMENT

Algorithm-centric prototype concept
- 0/1 Knapsack
- Dynamic Programming
- Constraint-aware selection
- Explainable output

NEXT IMPROVEMENTS

- Prerequisite checking
- Timetable conflict detection
- GPA-aware recommendations
- Preference ranking

SCALABILITY

- More elective groups
- Multiple constraints
- Faculty/admin portal
- Cloud deployment
- Analytics & reports

FINAL TAKEAWAY

A practical student decision can be transformed into an efficient, explainable optimization system.

THANK YOU